

GUIDE TO ATTACHMENT OPERATIONS

USE YOUR EQUIPMENT SAFELY AND EFFICIENTLY

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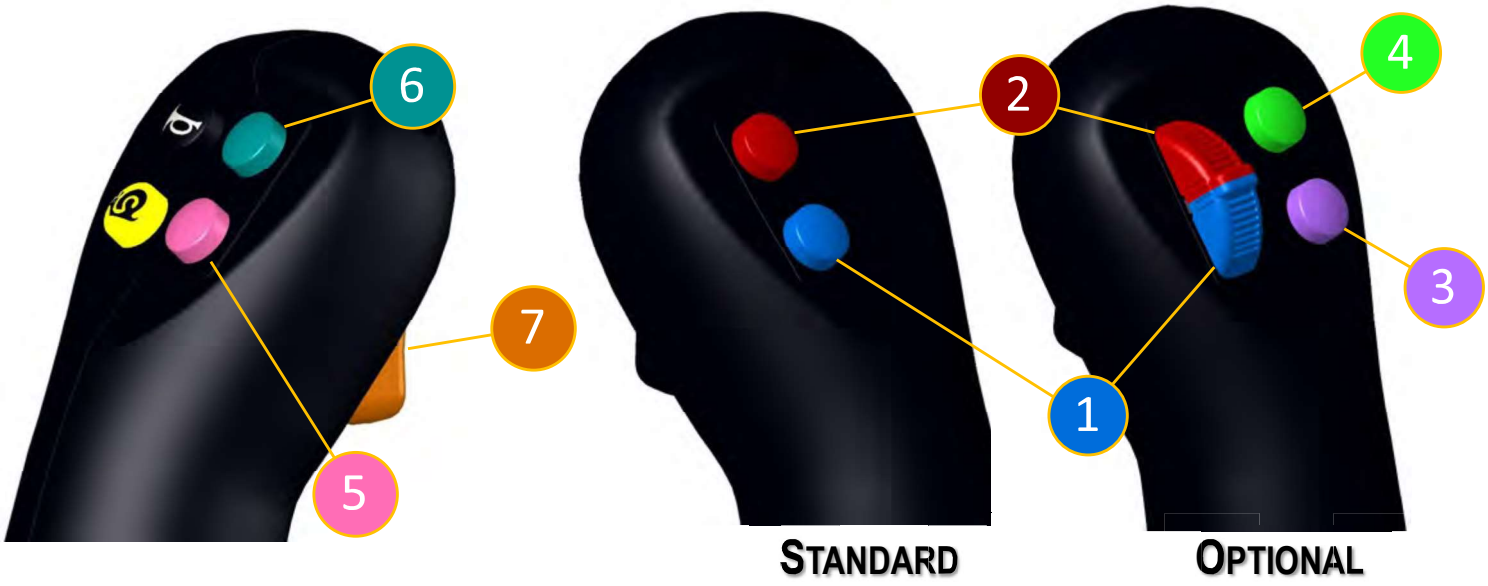
This operation guide is intended to provide an overview of joystick functionality for Cat® Skid Steer, Multi Terrain, and Compact Track Loaders in order to operate Cat attachments. The same information, along with more details on each attachment, is available in the attachment's Operation & Maintenance Manual (OMM). For more information, contact your local Cat dealer.

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ELECTRICAL WIRING

BUTTON	3	4	5	6	7
PIN	C	D	E	F	A

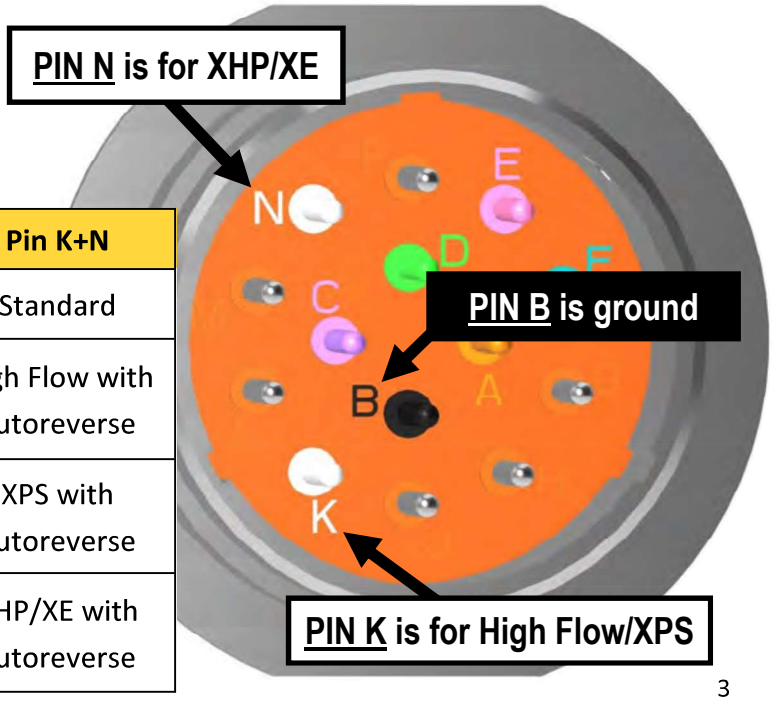


Optional Joystick required for certain attachments. Refer to individual attachment operation pages to see required controls.

The following table shows the flow given by connecting Pins K or N to Pin B depending on the flow capability of the machine. If neither Pin is connected, all machines give Standard Flow. All Attachments include harness/jumper plug if required for High Flow/XPS/XHP/XE.

Machine Type	Pin K	Pin N	Pin K+N
Standard Flow	Standard	Standard	Standard
High Flow	High Flow	Standard	High Flow with autoreverse
High Flow XPS	XPS	Standard	XPS with autoreverse
High Flow XHP/XE	XPS	XHP/XE with autoreverse	XHP/XE with autoreverse

MACHINE ELECTRICAL CONNECTOR FOR ATTACHMENTS



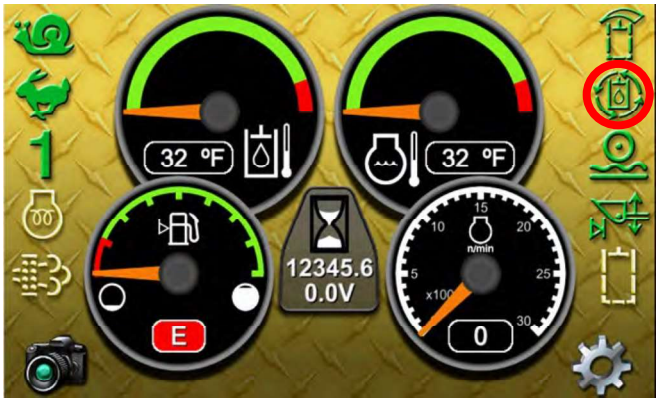
MACHINE FUNCTIONS

CONTINUOUS FLOW

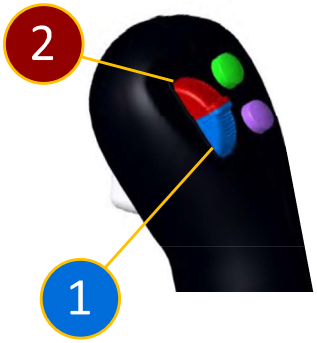
1. Press and release the Continuous Flow Switch, located on the upper right hand side switch panel.



2. Continuous Flow indicator will **FLASH** on the Advanced Machine Display.



3. Press and hold **2** on the right hand joystick for several seconds.



4. Continuous Flow indicator will appear **SOLID** on the Advanced Machine Display. Continuous flow is active, and button **2** can be released.

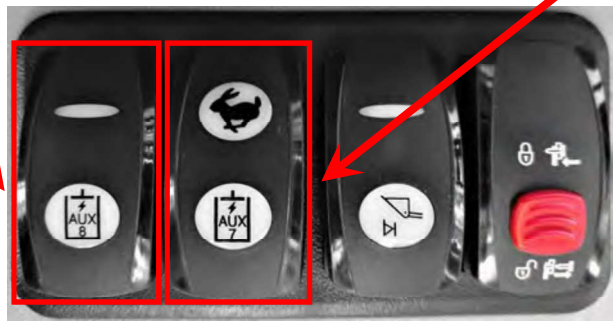
5. Press and release **2**, or the Continuous Flow Switch to deactivate continuous flow.

Aux Function 8

To turn on Aux 8, toggle the Aux 8 switch located on the upper left hand side switch panel. Used for certain attachments and available water kits, when equipped.

Aux Function 7

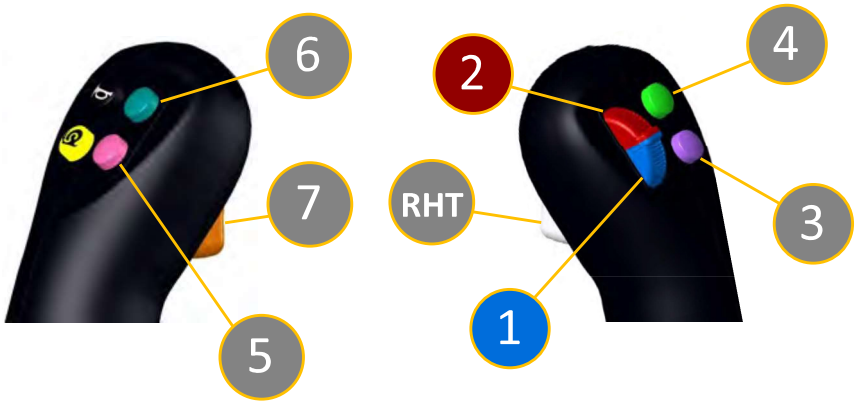
Press **DOWN** on the rocker switch, located on the left hand side switch panel, to select Aux Function 7. This enables the left hand joystick trigger to be used to control Attachments.



SMART DOZER BLADE

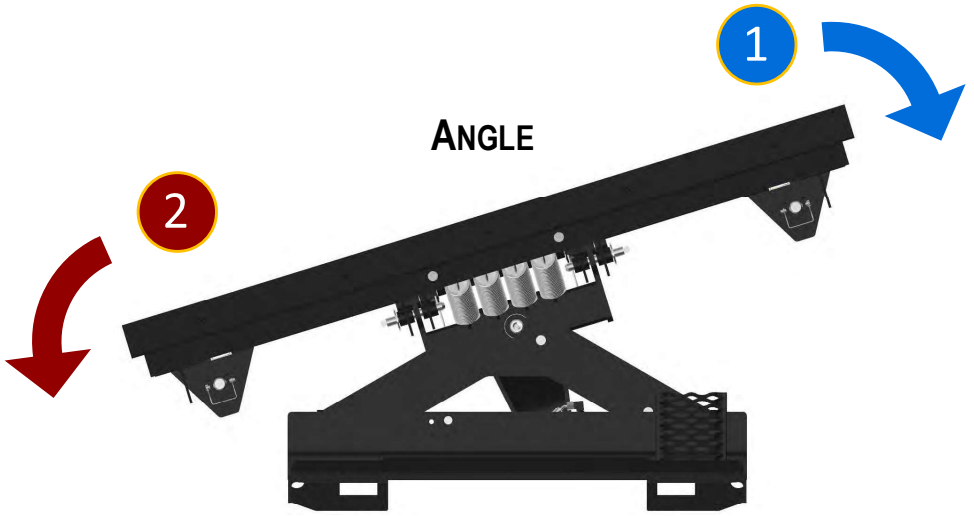
SMART DOZER BLADE

COMMAND	FUNCTION
1	Angle Right
2	Angle Left
Right Joystick—Right	Tilt Right
Right Joystick—Left	Tilt Left



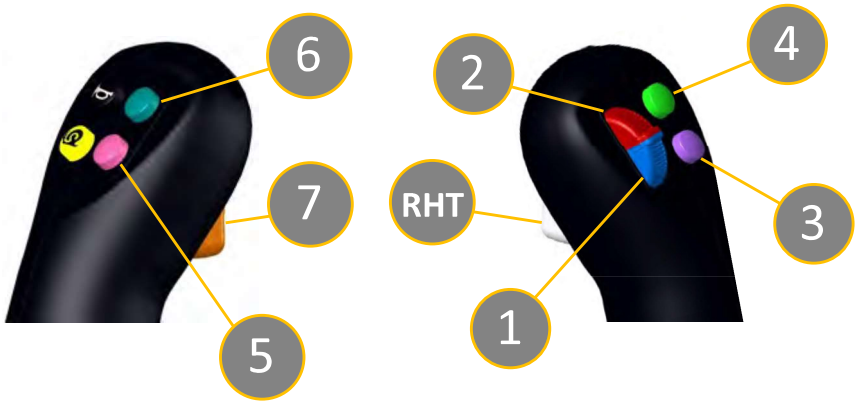
Only compatible with D3 series machines.





SMART DOZER BLADE

Only compatible with D3 series machines.



Many dozer blade functions are accessed through attachment control mode. To enter this mode:

1. Select the shovel icon on the machine display at start up to view the attachment control mode interface (1).
2. Select the shovel with cog icon to enter the attachment control mode settings.
3. On the top row of the settings, select either tilt or lift mode (see the following page for more information).
4. Press the button on the far right of the display to enter attachment control mode. Note that going in or out of this mode is only possible when the parking brake is active. The shovel icon will be green when in attachment



Select shovel with cog icon to enter Attachment Control Mode Settings.

(1) Attachment Control Mode Interface



(2) Home Screen—Green shovel icon indicates that attachment control mode is currently ON.



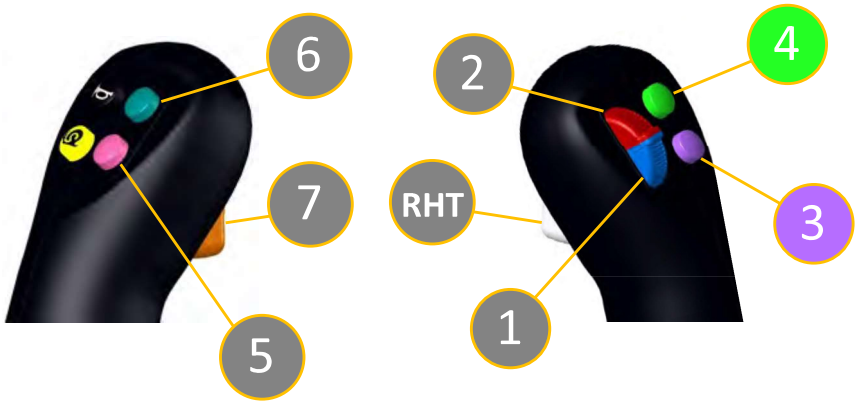
(3) Home Screen—White shovel icon indicates that attachment control mode is currently OFF.

SMART DOZER BLADE (CONTINUED)

SMART DOZER BLADE

Only compatible with D3 series machines.

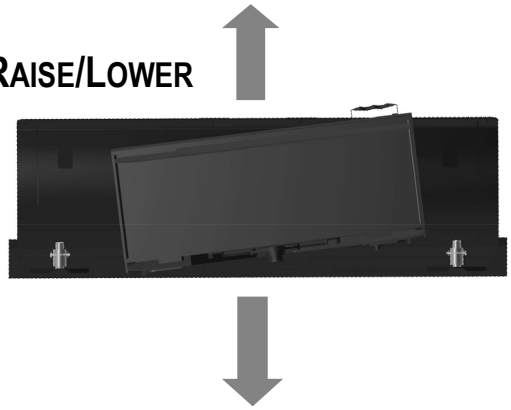
Note: The following machine functions are accessed through attachment control mode. See page 2 for instructions on how to enter this mode.



LIFT MODE

COMMAND	FUNCTION
Right Joystick—Forward	Lower Blade
Right Joystick—Back	Raise Blade
4	Pitch Forward
3	Pitch Back

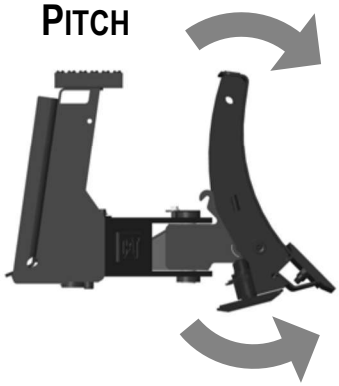
RAISE/LOWER



TILT MODE

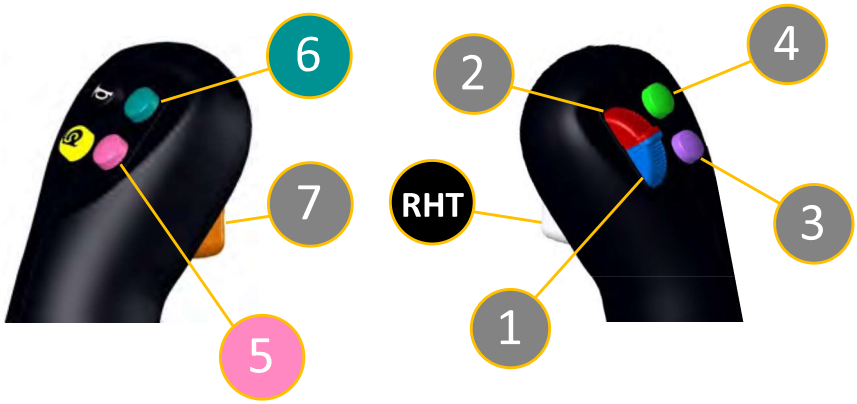
COMMAND	FUNCTION
Right Joystick—Forward	Pitch Forward
Right Joystick—Back	Pitch Back
4	Lower Blade
3	Raise Blade

PITCH



SMART DOZER BLADE

Note: The following machine functions are accessed through attachment control mode. See page 2 for instructions on how to enter this mode.



COMMAND	FUNCTION
RHT	Operator Assist
5	Memory/Recall/ Mirror
6	Return to Zero

Only compatible with D3 series machines.

Operator Assist—While in attachment control mode, press the right hand trigger (RHT) to activate Operator Assist. The gray box(es) in the display will turn green, and the green LED light on the attachment, visible from the operator's station, will illuminate when Operator Assist is active. Operator Assist is used to maintain a steady blade mainfall and/or blade slope during machine operation.

Memory/Recall/Mirror—This function can be used to program and recall a set blade slope position. Press and hold the memory button for 1 or more seconds to program the new position. Afterwards, the button can be pressed and released to return to the set position. The button can be pressed twice quickly to mirror the blade slope of the memory position. Operator Assist must be active to use this feature.

Return to Zero—Press and release this button to return to a blade slope of zero. Operator Assist must be active to use this feature.

Memory Position



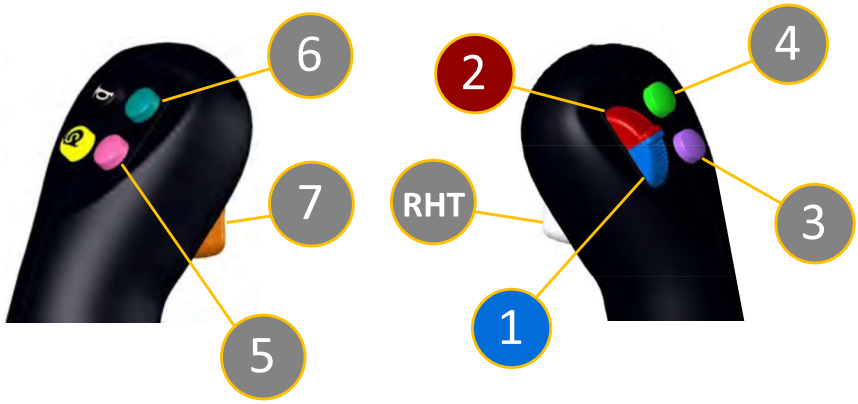
Mirror Memory Position

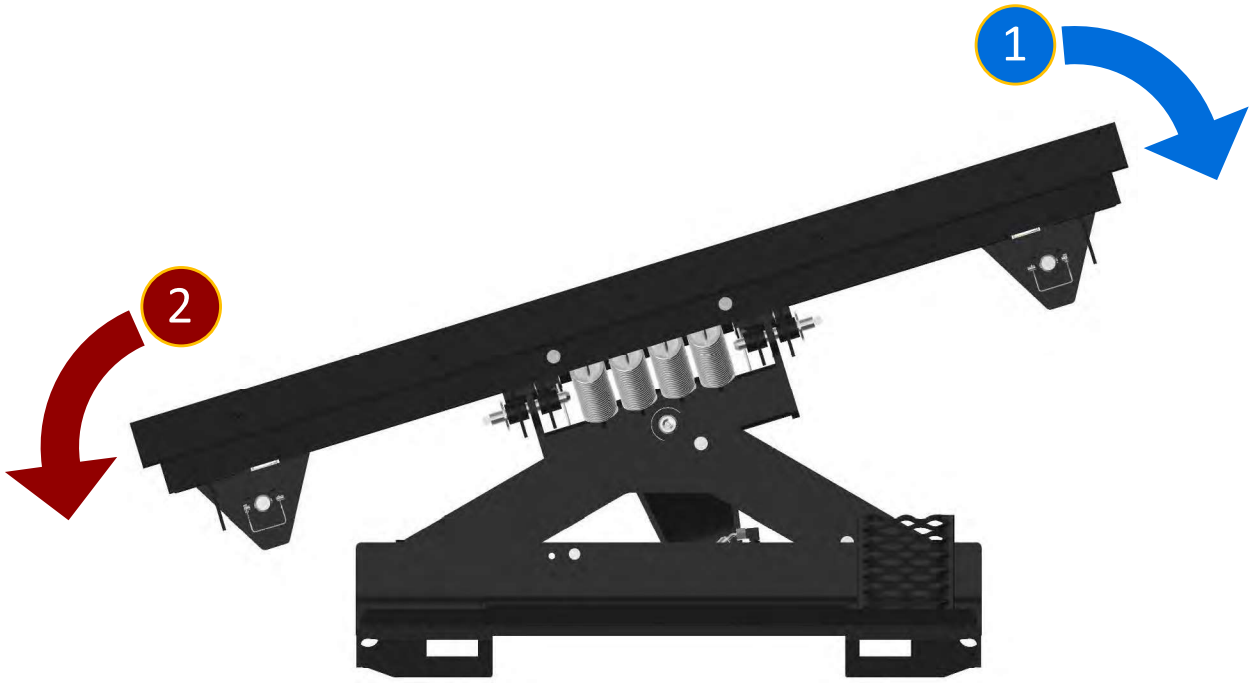


ANGLE & DOZER BLADES, SNOW PLOWS

ALL MODELS

BUTTONS	FUNCTION
1	Angle Right
2	Angle Left

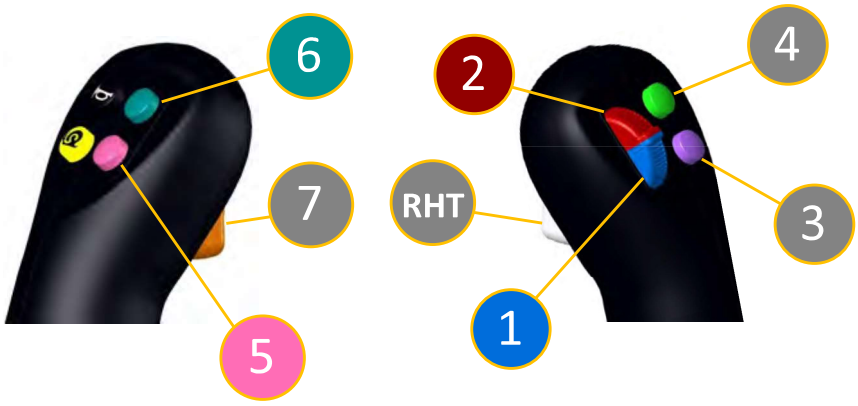




ANGLE & DOZER BLADES, SNOW PLOWS (CONT.)

DOZER BLADES

BUTTONS	FUNCTION
5+1	Tilt Right
5+2	Tilt Left
6+1	Tilt + Angle Right
6+2	Tilt + Angle Left



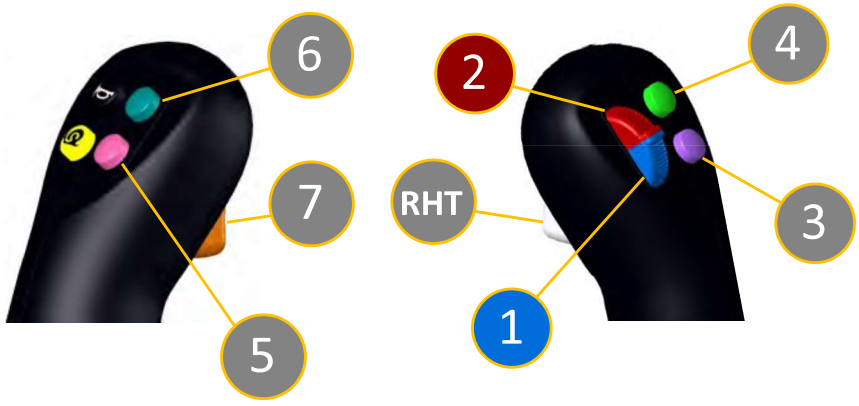




AUGERS

ALL MODELS

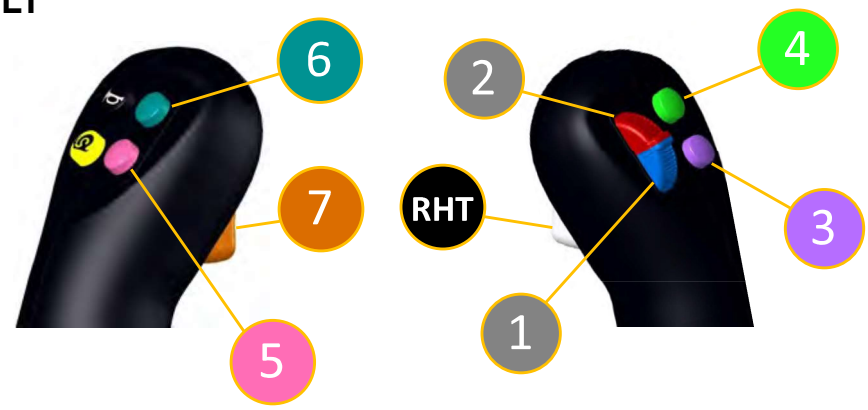
BUTTONS	FUNCTION
1	Counterclockwise
2	Clockwise





BH130—WITH D3 SERIES ONLY

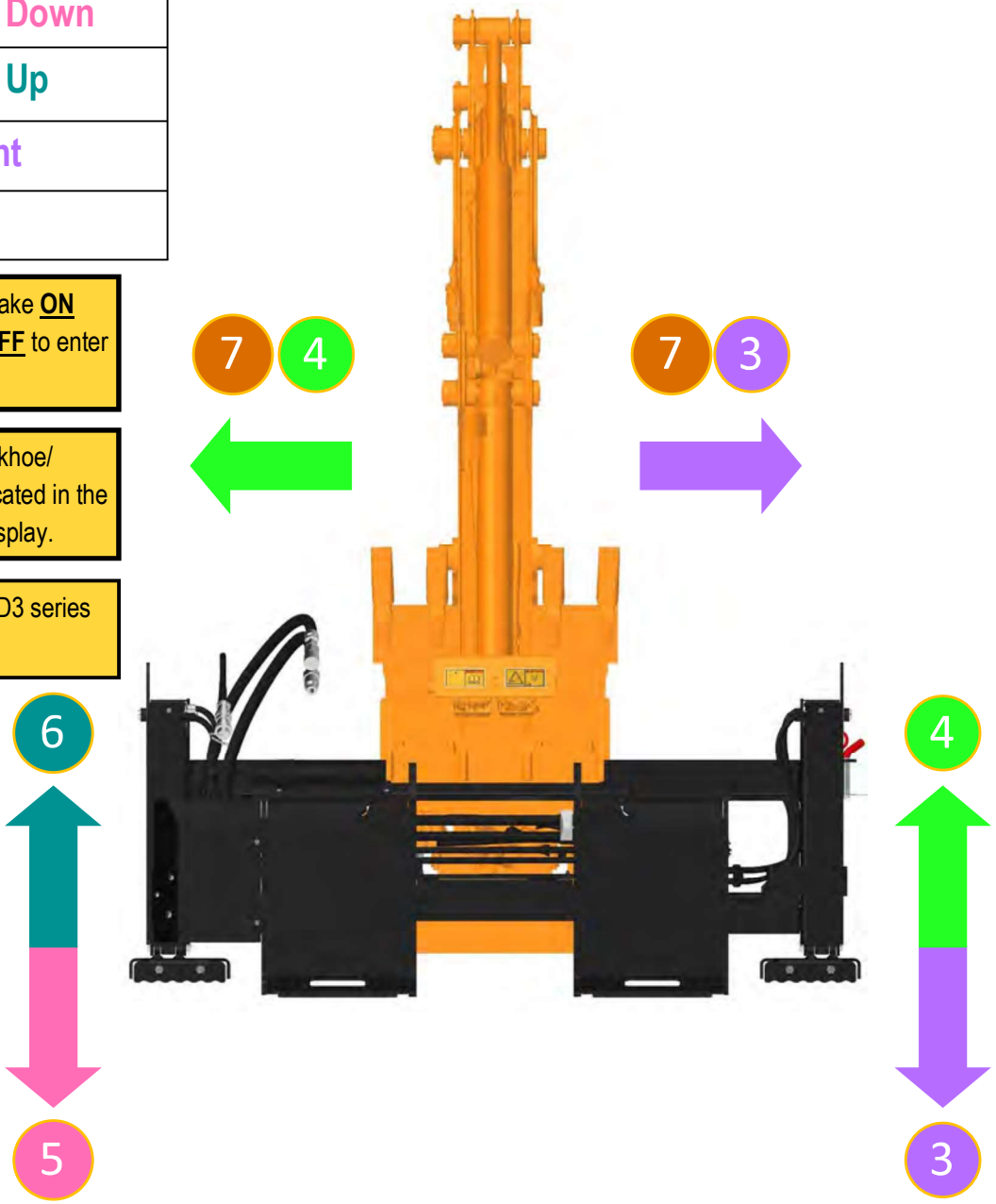
BUTTONS	FUNCTION
RH Trigger	Toggle Attachment Mode
3	Right Stab Down
4	Right Stab Up
5	Left Stab Down
6	Left Stab Up
7+3	Shift Right
7+4	Shift Left



Must have Parking Brake **ON** and Hydraulic Lock **OFF** to enter Attachment Mode.

Pattern select for Backhoe/ Excavator controls located in the advanced machine display.

Only compatible with D3 series machines.

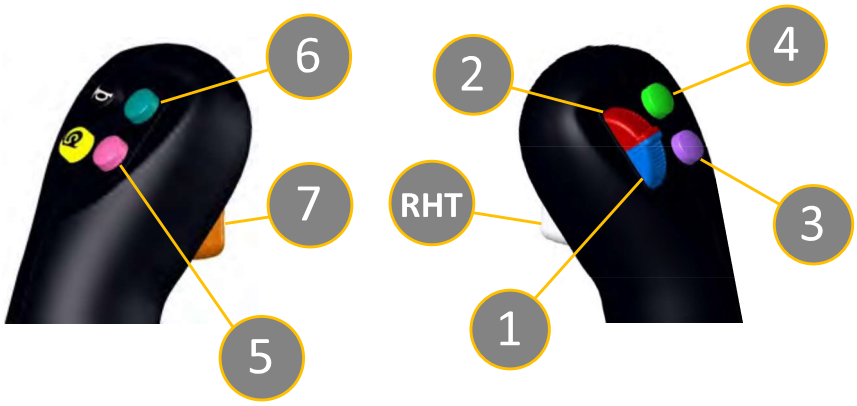


BH130—WITH D3 SERIES ONLY

Must have Parking Brake **ON** and Hydraulic Lock **OFF** to enter Attachment Mode.

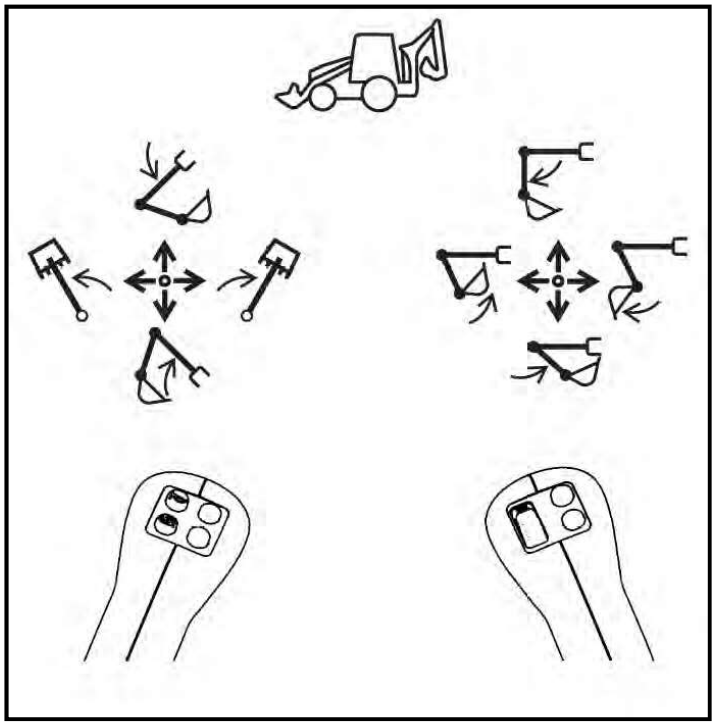
Pattern select for Backhoe/ Excavator controls located in the advanced machine display.

Only compatible with D3 series machines.

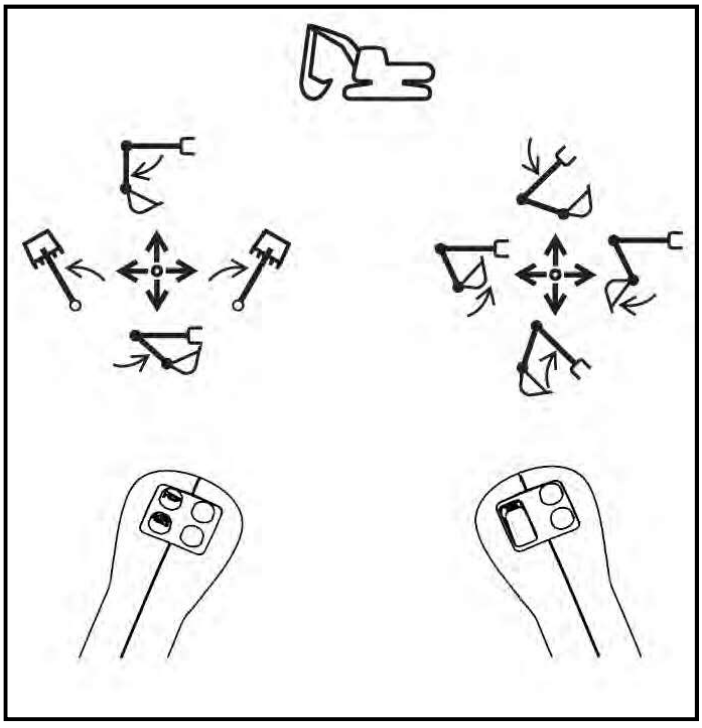


MAIN ATTACHMENT MODE FUNCTIONS

BACKHOE MODE



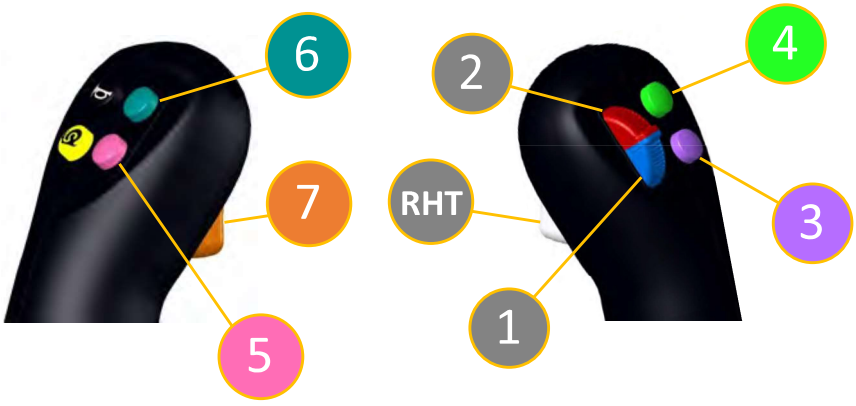
EXCAVATOR MODE



BOX BLADES

ALL MODELS—MANUAL MODE

BUTTONS	FUNCTION
3	Right Side Down
4	Right Side Up
5	Left Side Down
6	Left Side Up
7	Toggle Auto/ Manual



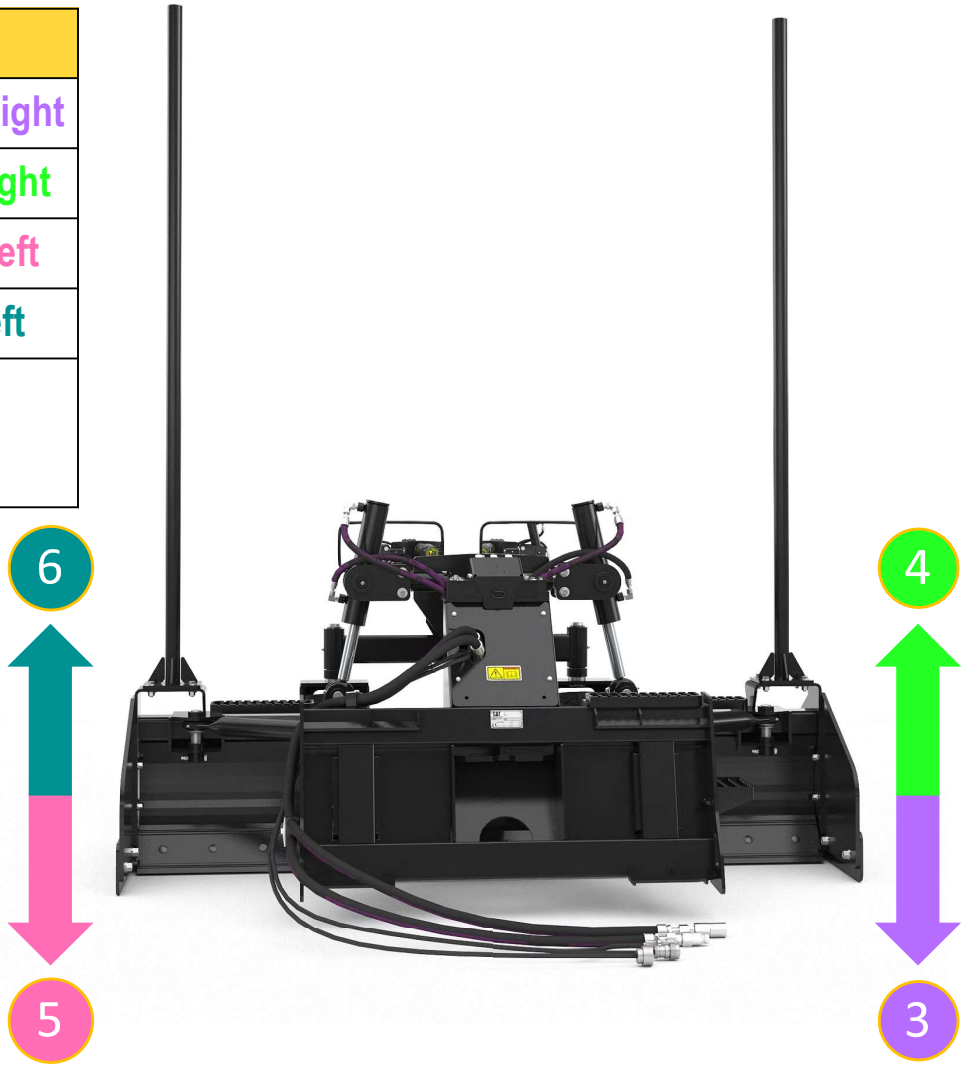
ALL MODELS—AUTO MODE

BUTTONS	FUNCTION
3	Decrement Right
4	Increment Right
5	Decrement Left
6	Increment Left
7	Toggle Auto/ Manual

Must be in Continuous Flow.

Must have Aux 7 enabled to allow toggle between auto and manual modes.

Must have Aux 8 enabled to provide power to the box blade.



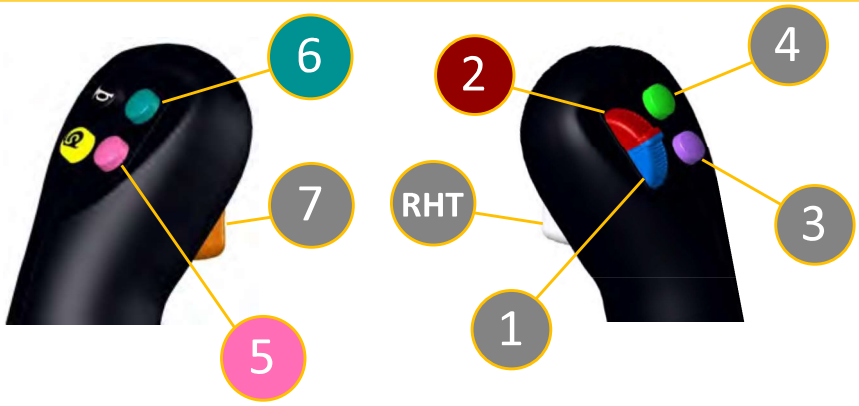
BROOMS, ANGLE

ALL MODELS

BUTTONS	FUNCTION
2	Broom Rotation

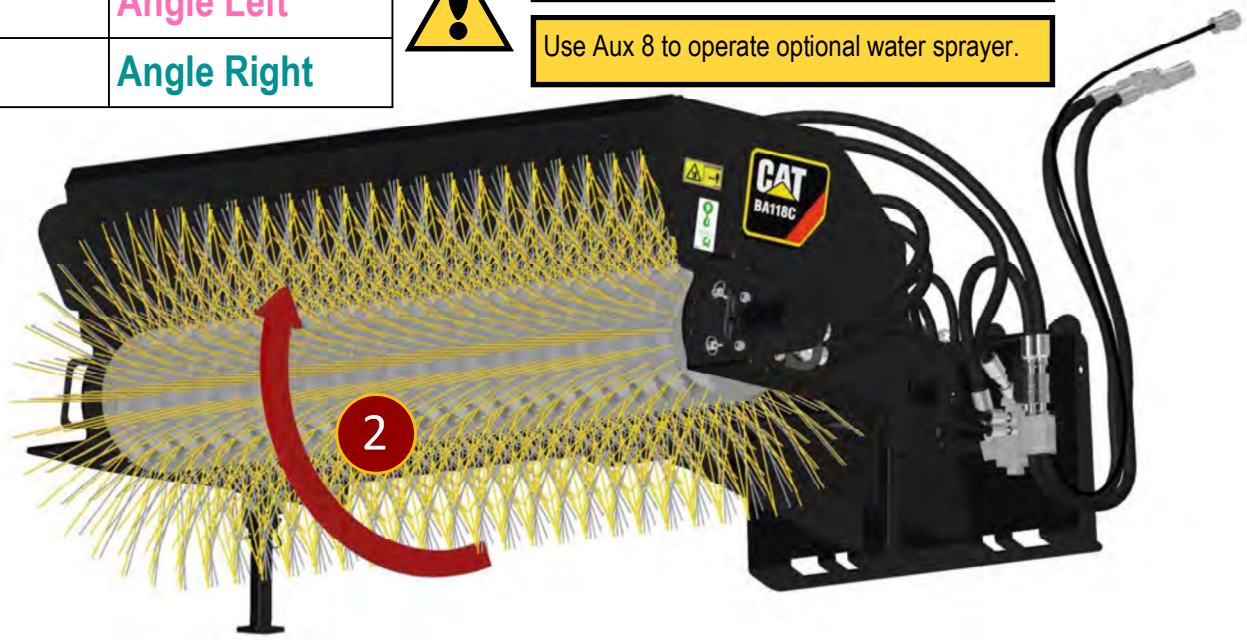
MODELS WITH HYDRAULIC ANGLE

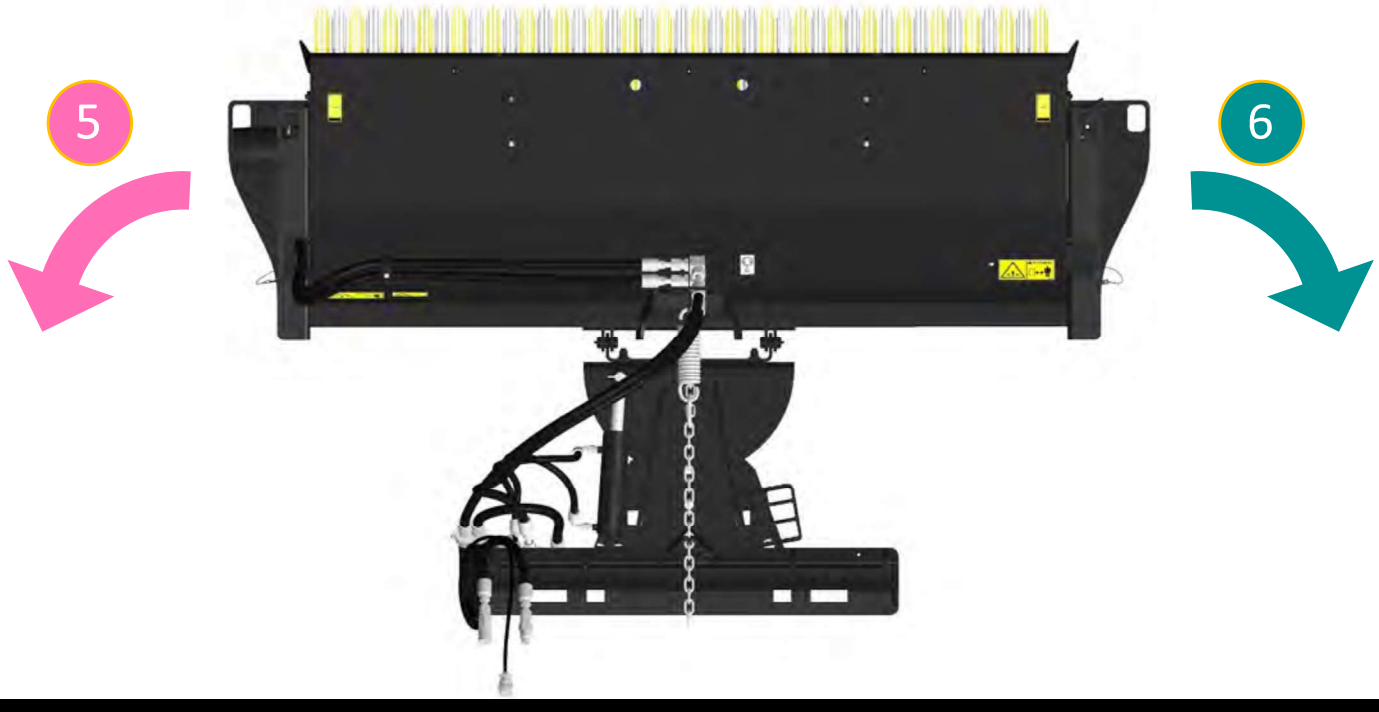
BUTTONS	FUNCTION
5	Angle Left
6	Angle Right



Must be in Continuous Flow.

Use Aux 8 to operate optional water sprayer.



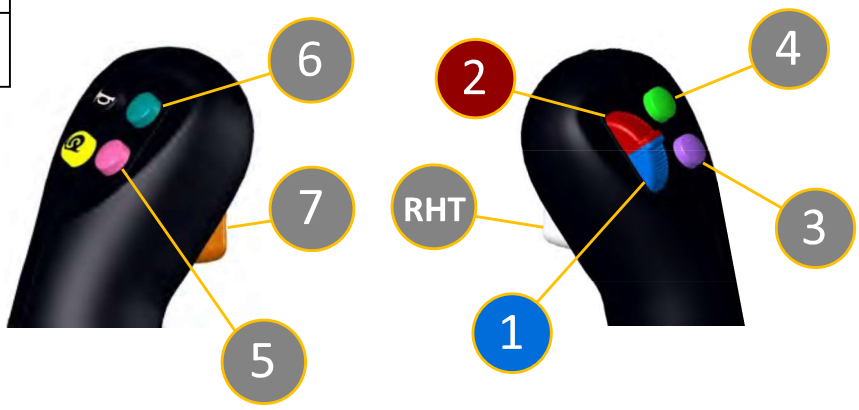


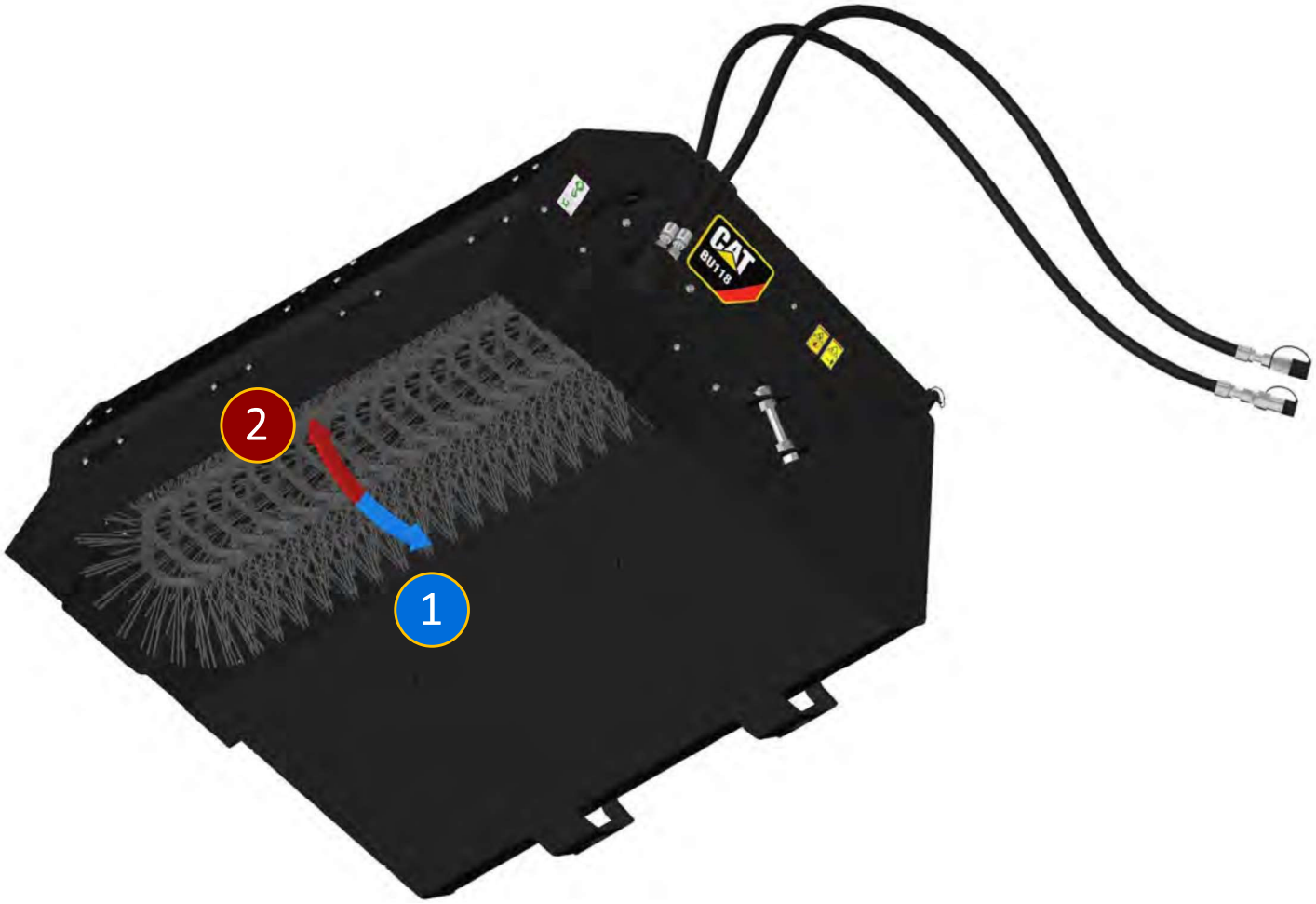
BROOMS, PICKUP AND UTILITY

ALL MODELS

BUTTONS	FUNCTION
1	Dust Pan Rotation
2	Over Brush Rotation

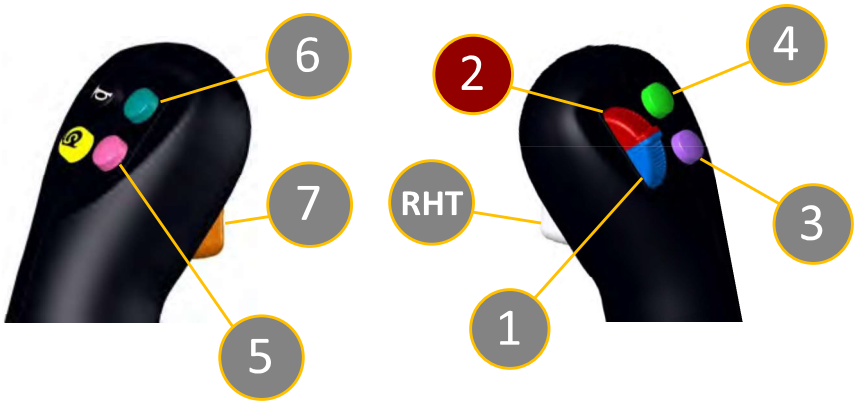
Use Aux 8 to operate optional water sprayer.

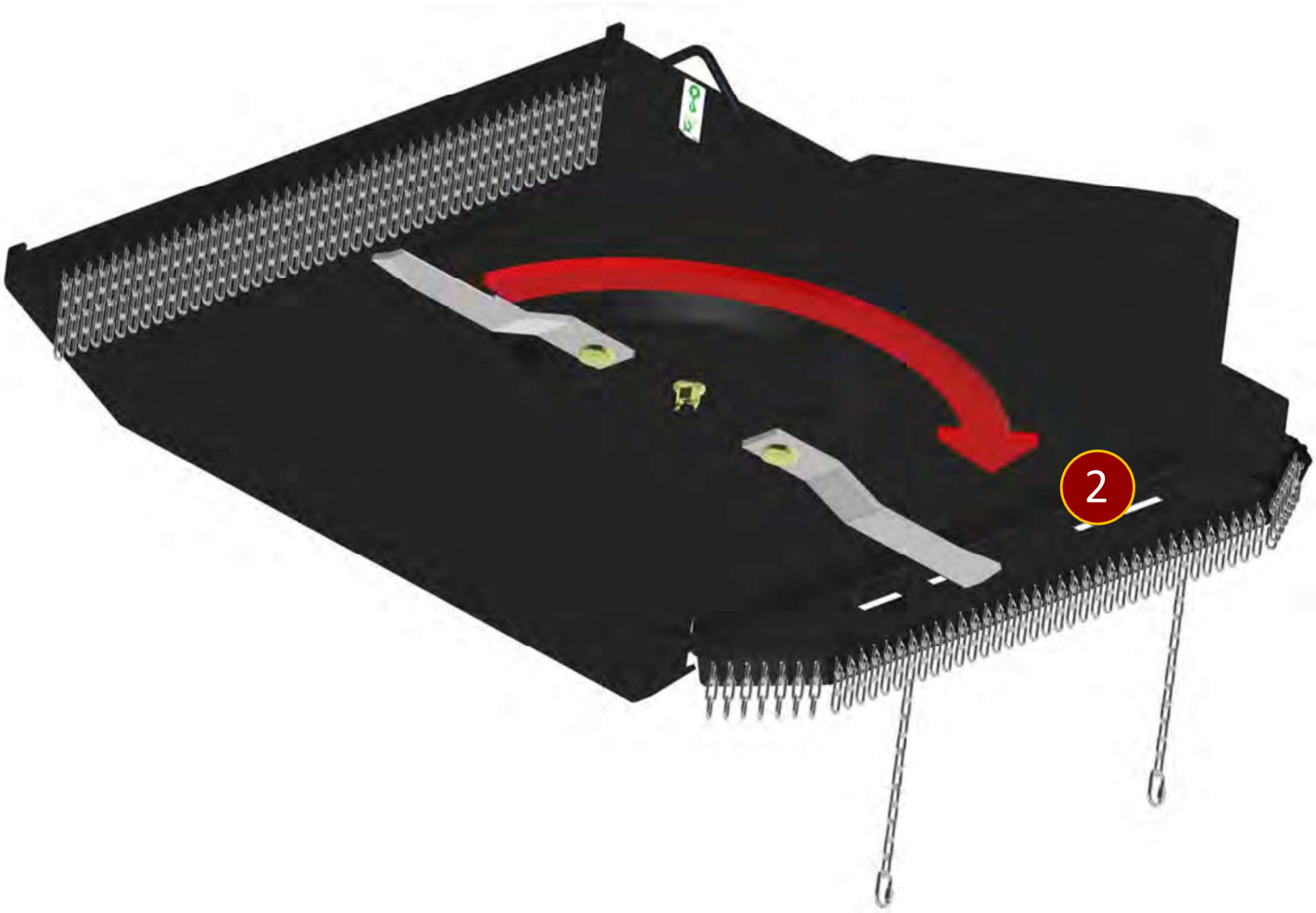




ALL MODELS

BUTTONS	FUNCTION
2	Forward



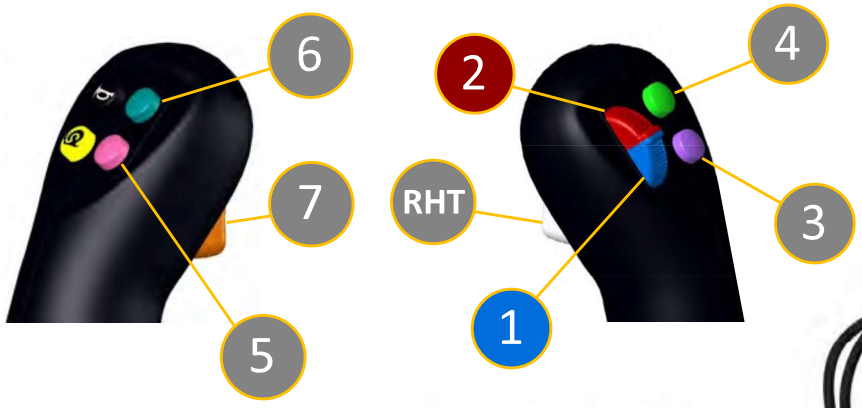


All Models

Buttons	Function
2	Drum Rotation

PC104B, PC205B

Buttons	Function
1	Shift Right
2	Shift Left



Use Aux 8 to operate optional water sprayer.





On PC104B and PC205B models, use this valve to switch to side shift function.

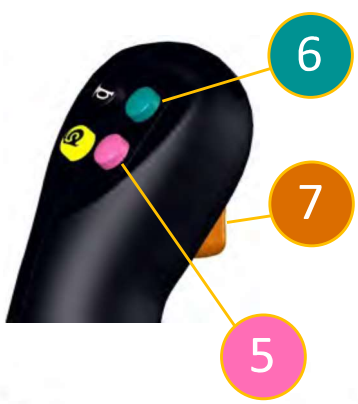


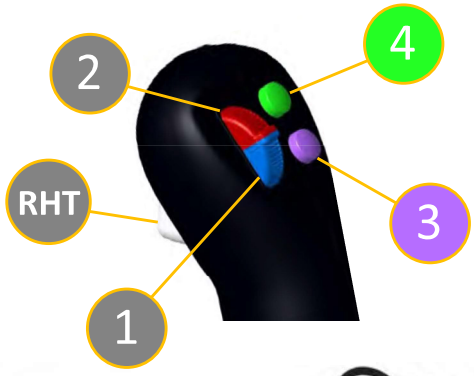


COLD PLANERS (CONT.)

PC305B, PC306B, PC310B, PC408B, PC412B

BUTTONS	FUNCTION
3	Shift Left
4	Shift Right
5+3	Left Skid Down
5+4	Left Skid Up
6+3	Right Skid Down
6+4	Right Skid Up
7+3	Tilt Drum Left
7+4	Tilt Drum Right

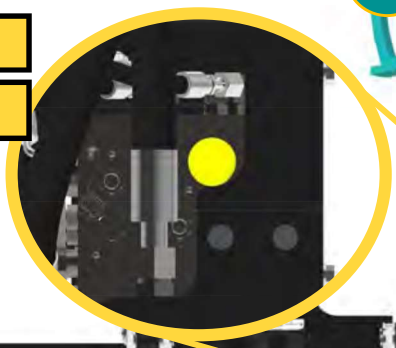




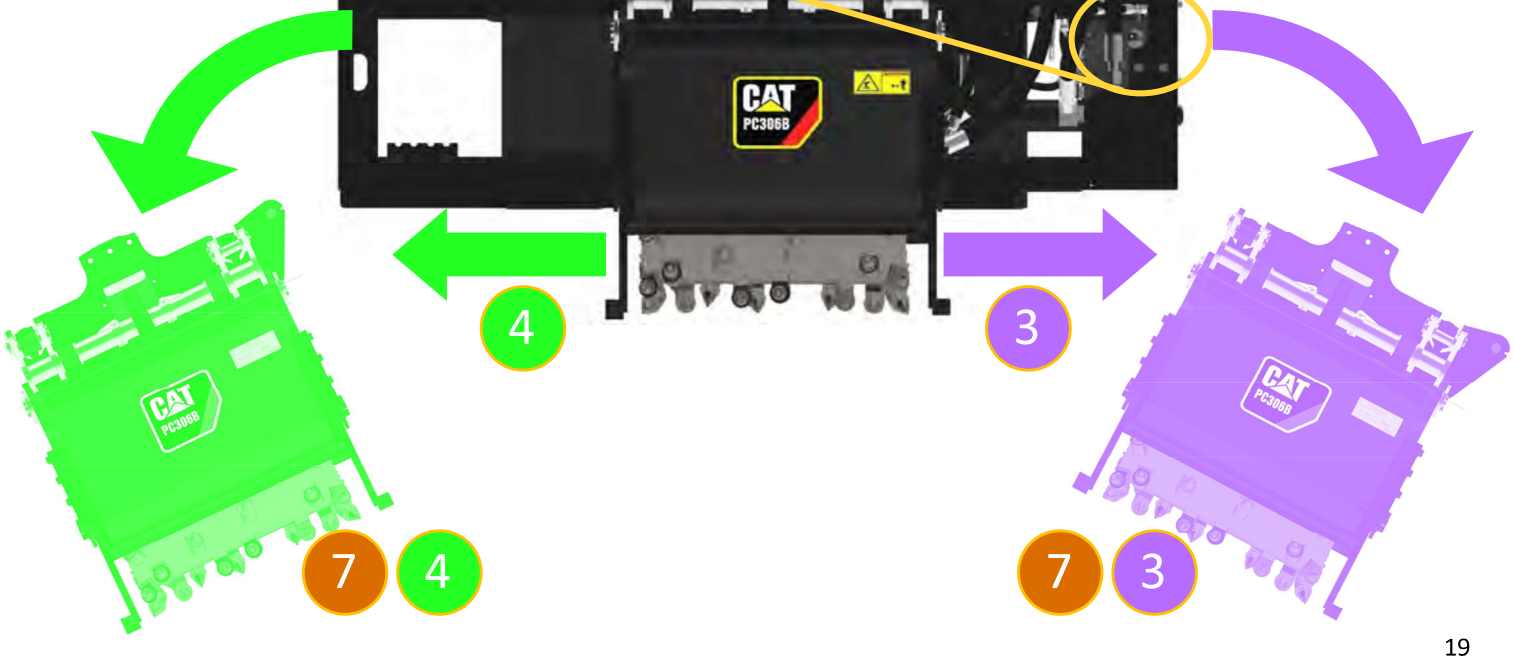


Must be in Continuous Flow.

Must have Aux 7 enabled.

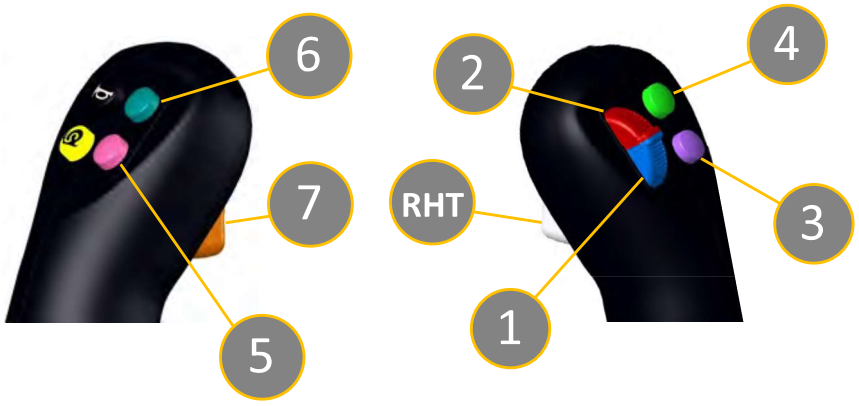


Use this valve to switch between fixed and floating head tilt.



SMART GRADER BLADE

Only compatible with D3 series machines.



Many Grader blade functions are accessed through attachment control mode. To enter this mode:

1. Select the shovel icon on the machine display at start up to view the attachment control mode interface (1).

2. Press the button on the far right of the display to enter attachment control mode. Note that going in or out of this mode is only possible when the parking brake is active. The shovel icon will be green when in attachment control mode (2), and will be white when in normal skid steer mode (3).



(1) Attachment Control Mode Interface



(2) Home Screen—Green shovel icon indicates that attachment control mode is currently ON.



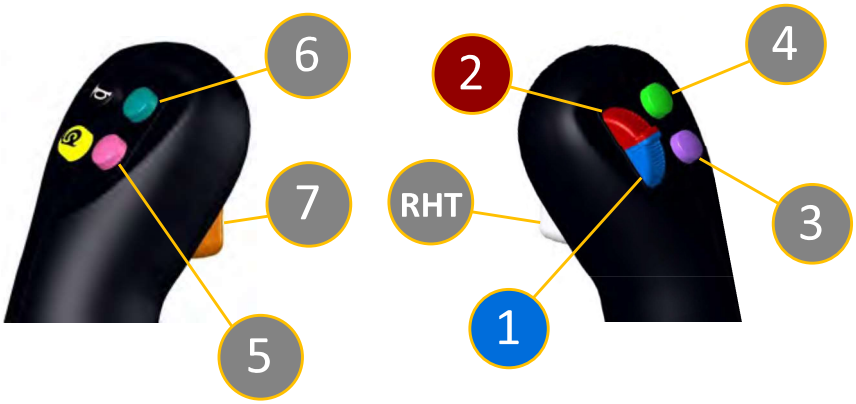
(3) Home Screen—White shovel icon indicates that attachment control mode is currently OFF.

SMART GRADER BLADE (CONTINUED)

SMART GRADER BLADE

COMMAND	FUNCTION
1	Angle Right
2	Angle Left
Right Joystick—Right	Tilt Right
Right Joystick—Left	Tilt Left
Right Joystick—Forward	Lower Blade
Right Joystick—Back	Raise Blade

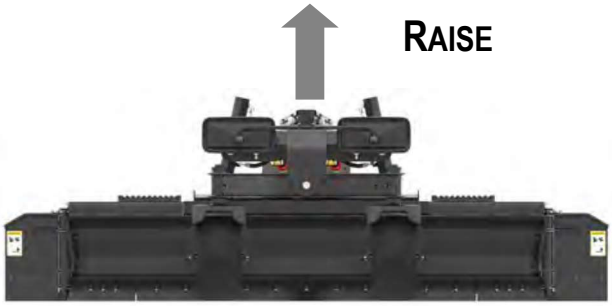
Only compatible with D3 series machines.



TILT

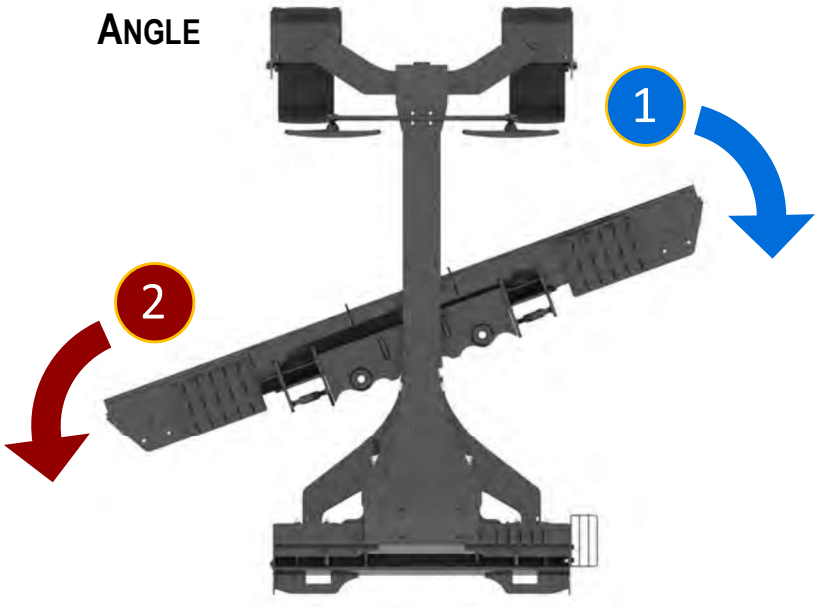


RAISE



LOWER

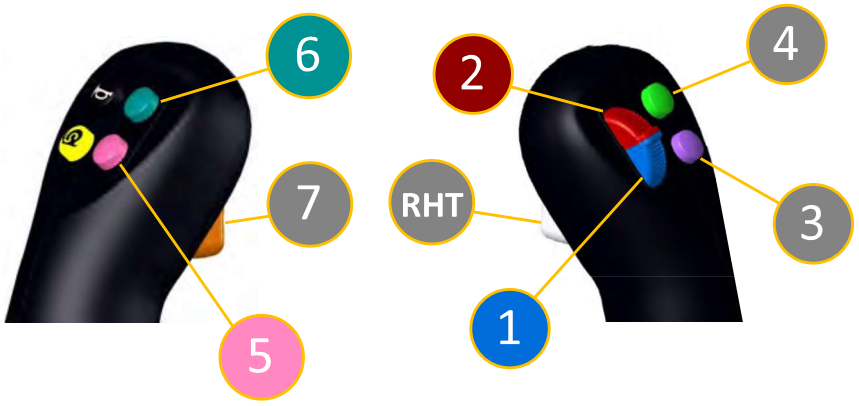
ANGLE



SMART GRADER BLADE (CONTINUED)

SMART GRADER BLADE

COMMAND	FUNCTION
6+1	Right Wing-Rotate Back
6+2	Right Wing-Rotate Forward
5+1	Left Wing-Rotate Back
5+2	Left Wing-Rotate Forward



Only compatible with D3 series machines.

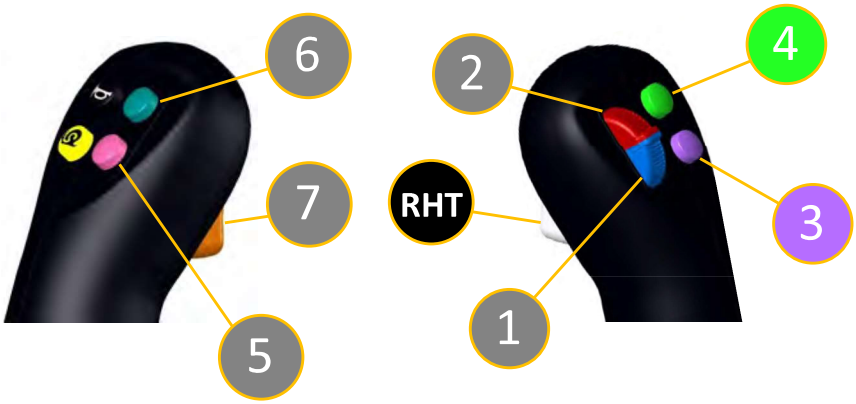
WING ROTATION



SMART GRADER BLADE (CONTINUED)

SMART GRADER BLADE

Note: The following machine functions are accessed through attachment control mode. See page 2 for instructions on how to enter this mode.



COMMAND	FUNCTION
3	Memory/Recall/ Mirror
4	Return to Zero

Only compatible with D3 series machines.

Memory/Recall/Mirror—This function can be used to program and recall a set blade slope position. Press and hold the memory button for 1 or more seconds to program the new position. Afterwards, the button can be pressed and released to return to the set position. The button can be pressed twice quickly to mirror the blade slope of the memory position. Operator Assist must be active to use this feature.

Return to Zero—Press and release this button to return to a blade slope of zero. Operator Assist must be active to use this feature.

Memory Position



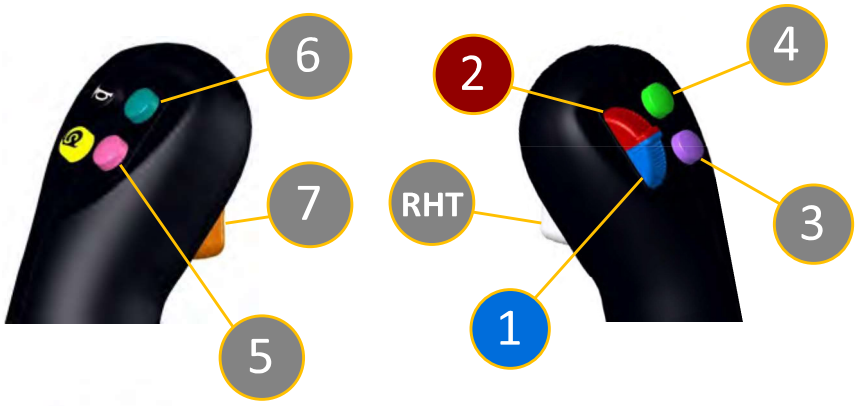
Mirror Memory Position



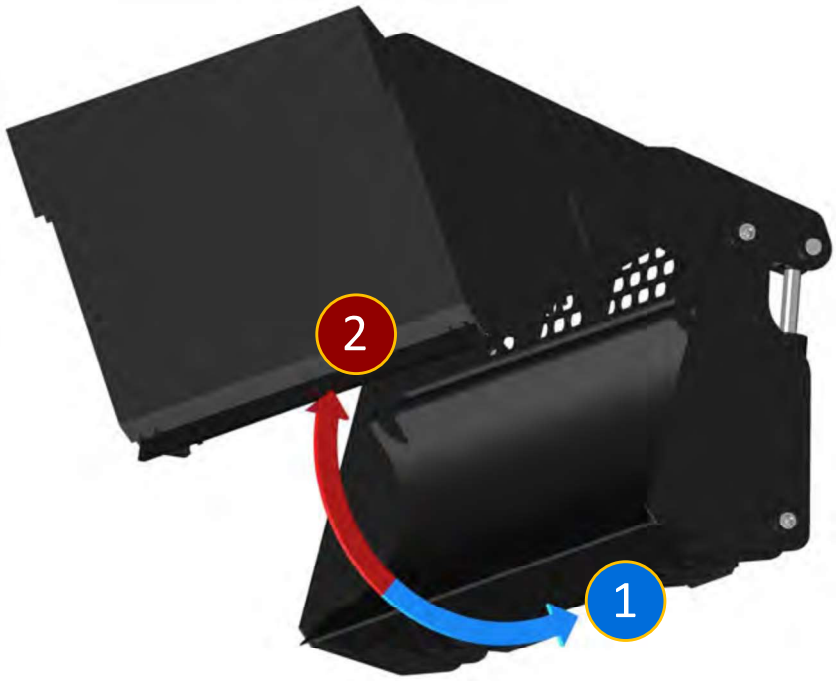
GRAPPLES, MULTI-PURPOSE BUCKETS

ALL MODELS

BUTTONS	FUNCTION
1	Close
2	Open







ALL MODELS

BUTTONS	FUNCTION
2	Hammer

